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Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the left sublist
b. At the beginning of the list
c. In the middle of the list
d. In the right sublist ✓

Question **2**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It can be applied only if the list is sorted ✓
b. It works on unsorted lists
c. It always starts from the beginning of the list
d. It compares elements sequentially

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Bubble search
b. Quick search
c. Linear search ✓
d. Merge search

Question **4**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

- a. Compare the target element with the middle element in the list ✓
- b. Compare the target element with the first element in the list
- c. Divide the list into two equal parts
- d. Sort the list

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

- a. When the list is unsorted
- b. When the list is very small
- c. When the list contains duplicate elements
- d. When the list is sorted ✓

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

- a. The list must be sorted
- b. There must be a mechanism to delete and/or insert elements in the list ✓
- c. There should be direct access to the middle element in any sublist
- d. Number values should only be present

Number values should only be present

Question **7**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O(n log n)
- b. O(log n) ✓
- c. O
- d. O(1)

Question **8**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(n log n)
b. O(1)
c. O(log n)
d. O👎 ✓

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Sorting the list before searching
b. Dividing the list in half repeatedly
c. Skipping every second element
d. Checking each element sequentially ✓

Question **10**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and unsorted
b. When the list is large and sorted
c. When the list is small and unsorted ✓
d. When the list is small and sorted

Question **11**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Linear search ✓
b. Hash search
c. Binary search
d. Both (1) & (3)

Question **12**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 100
b. 1000
c. 500
d. 10 ✓

Question **13**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. Item is the last element in the array or item is not there at all
b. When the item is not the array at all
c. When the item is the last element in the array
d. When the item is somewhere in the middle of the array ✓

Question **14**

Incorrect

Mark 0.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Binary search
b. Hash search ✗
c. Interpolation search
d. Linear search

Question **15**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the element is found in the list ✓
b. When the list contains duplicate elements
c. When the search algorithm finishes
d. When the list is sorted